

Set Notation and Venn Diagrams

A set is a list in curly brackets. Four symbols ($\cap$, $\cup$, $'$, $\in$) and one rule for Venn diagrams: fill from the middle outwards.



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THE TOPIC ON ONE MAP

Set Notation and Venn Diagrams

01 Set notation: list the members

$A \cap B$, $A \cup B$, A'

"is $24 \in A$?"

"one possible set C "

02 Read a Venn diagram

list the members

$n(A \cup B)$

$n(A' \cap B')$

03 Shade a region

shade $A \cap C'$

describe the shaded region

04 Fill it in: two sets

50 people: cat, dog

$n(A)$, $n(A \cap B)$ given

05 Three sets with x

dog, cat, rabbit

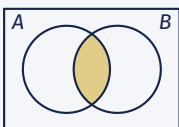
total = 80: find x

$3x$, $x + 2$ in regions

THE FOUR SYMBOLS Each one is a region — see it before you list it:

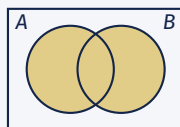
ZONE $\cap$

$A \cap B$: in both



ZONE $\cup$

$A \cup B$: in either



ZONE $'$

A' : not in A



ZONE $\cap'$

$A \cap B'$: A only



01 Set Notation: List the Members

Write the universal set E out in full first. Then test every member against the rule. Each member is listed **once**.

WHEN YOU SEE → YOU DO

$A \cap B$ (“and”)	→ members in both
$A \cup B$ (“or”)	→ members in either — list each once
A'	→ everything in E that is not in A
$\emptyset$, “ $B \cap K = \emptyset$ ”	→ the empty set: nothing in common
“is it true that $24 \in A$?”	→ check E first — 24 may not even be in it
“list the members of one possible set C ”	→ build it from the conditions, then check each one

Write out E Write out A and B

Apply the symbol

Check each member

WORKED MODEL • JANUARY 2021 P1H

$E = \{20, 21, \dots, 29\}$, $A = \{\text{odd numbers}\}$, $B = \{\text{multiples of 3}\}$. List $A \cap B$ and $A \cup B$.

1 $A = \{21, 23, 25, 27, 29\}$, $B = \{21, 24, 27\}$

write them out

2 $A \cap B = \{21, 27\}$

in both

ANSWER $A \cup B = \{21, 23, 24, 25, 27, 29\}$

× **TRAP** Writing 21 twice in $A \cup B$, or forgetting the curly brackets $\{ \}$.

IDEA 01

Practise: Set Notation: List the Members

My questions first, then selected practice, then real past papers.

Q1 DR ESLAM Both and either

CH 1 • TOPIC 2 2 marks

$E = \{1, 2, \dots, 12\}$, $A = \{\text{factors of } 12\}$, $B = \{\text{even numbers}\}$. List (a) $A \cap B$ (b) A' .

SELECTED PRACTICE

Q2**PRACTICE** Practice • Easy

CH 1 • TOPIC 2 3 marks

1 (a) $A = \{p, r, a, g, u, e\}$

$B = \{p, a, r, i, s\}$

$C = \{b, u, d, a, p, e, s, t\}$

List the members of the set

i) $A \cap B$,

[1]

ii) $B \cup C$.

[1]

(2 marks)

(b) $D = \{r, o, m, e\}$

$E = \{l, i, s, b, o, n\}$

$F = \{b, e, r, l, i, n\}$

Put one of the letters D , E or F in the box below to make the statement correct.

$$A \cap \square = \emptyset$$

Explain your answer.

(1 mark)

Q3**PRACTICE**

Practice • Easy

CH 1 • TOPIC 2 4 marks

11 (a) $\mathcal{E} = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$
 $A = \{2, 3, 5, 7\}$
 $B = \{4, 6, 8, 10\}$

Explain why $A \cap B = \emptyset$

(1 mark)

(b) $x \in \mathcal{E}$ and $x \notin A \cup B$

Write down the **two** possible values of x .

(1 mark)

(c) Set C is such that

$$A \cup B \cup C = \mathcal{E}$$

$$A \cap C = \{2\}$$

$$B \cap C = \{4, 6, 10\}$$

List all the members of set C .

(2 marks)

Q4**PRACTICE**

Practice • Medium

CH 1 • TOPIC 2**2 marks**

7 $\xi = \{\text{whole numbers}\}$

$A = \{\text{factors of } 100\}$

$B = \{\text{multiples of } 5\}$

List the members of the set $A \cap B$.

(2 marks)

Q5**PRACTICE**

Practice • Hard

CH 1 • TOPIC 2 3 marks

4 (a) $\xi = \{\text{positive whole numbers less than } 13\}$

$A = \{\text{even numbers}\}$

$B = \{\text{multiples of } 3\}$

$C = \{\text{prime numbers}\}$

List the members of the set

i) $A' \cap B'$,

ii) $B \cup C$.

(2 marks)

(b) Is it true that $14 \in A$?

Tick (✓) the appropriate box.

☐ Yes ☐ No

Explain your answer.

(1 mark)

PAST PAPERS

Q6**PAST PAPER**

Edexcel 4MA1 • January 2020 • Paper 1HR • Q4

CH 1 • TOPIC 2 3 marks

4 $B = \{b, l, u, e\}$

$G = \{g, r, e, y\}$

$W = \{w, h, i, t, e\}$

(a) List all the members of the set

(i) $B \cup G$

.....

(ii) $W \cap G'$

.....

Serena writes down the statement $B \cap G \cap W = \emptyset$

(b) Is Serena's statement correct?

You must give a reason for your answer.

.....

.....

.....

(Total for Question 4 is 3 marks)

Q7**PAST PAPER**

Edexcel 4MA1 • January 2021 • Paper 1H • Q4

CH 1 • TOPIC 2 2 marks

4 $\mathcal{E} = \{20, 21, 22, 23, 24, 25, 26, 27, 28, 29\}$

$$A = \{\text{odd numbers}\}$$

$$B = \{\text{multiples of 3}\}$$

List the members of the set

(i) $A \cap B$

.....

(ii) $A \cup B$

.....

(Total for Question 4 is 2 marks)

Q8**PAST PAPER**

Edexcel 4MA1 • January 2021 • Paper 2HR • Q3

CH 1 • TOPIC 2 3 marks

3 $\mathcal{E} = \{21, 22, 23, 24, 25, 26, 27, 28, 29, 30\}$

$$A = \{22, 24, 26, 28, 30\}$$

$$B = \{21, 24, 27, 30\}$$

(a) List the members of the set

(i) $A \cap B$

.....

(ii) A'

.....

$$C = \{23, 25, 29\}$$

(b) Using set notation, find an expression for C in terms of A and B .

.....

(Total for Question 3 is 3 marks)

.....

Q9**PAST PAPER**

Edexcel 4MA1 • May 2021 • Paper 2H • Q4

CH 1 • TOPIC 2 3 marks**4** $\mathcal{E} = \{\text{letters of the alphabet}\}$

$$B = \{b, r, a, z, i, l\}$$

$$I = \{i, r, e, l, a, n, d\}$$

(a) List the members of the set

(i) $B \cup I$

.....

(ii) $B \cap I'$

.....

$$K = \{k, e, n, y, a\}$$

Cody writes down the statement $B \cap K = \emptyset$

Cody's statement is wrong.

(b) Explain why.

.....

(Total for Question 4 is 3 marks)

.....

Q10**PAST PAPER**

Edexcel 4MA1 • June 2022 • Paper 2HR • Q4

CH 1 • TOPIC 2 5 marks

4 $\mathcal{E} = \{9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20\}$

$A = \{\text{multiples of 3}\}$

$B = \{\text{odd numbers}\}$

(a) List the members of the set

(i) $A \cap B$

.....

(ii) $A \cup B$

.....

(b) Is it true that $24 \in A$?

Tick one of the boxes below.

Yes

No

☐
☐

Give a reason for your answer.

.....

Set C has 4 members such that $C \cap B' = \{10, 18\}$

(c) List the members of one possible set C

.....

(Total for Question 4 is 5 marks)

Q11**PAST PAPER**

Edexcel 4MA1 • June 2024 • Paper 2HR • Q4

CH 1 • TOPIC 2 5 marks

4 $\mathcal{E} = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$

$A = \{\text{factors of } 6\}$

$B = \{\text{prime numbers}\}$

(a) List the members of the set

(i) $A \cup B$

(ii) A'

Harpreet states that $A \cap B = \emptyset$

Harpreet is incorrect.

(b) Explain why.

C is a set with 4 members such that

the set $A \cap C$ has 2 members

the set $B \cap C$ has 2 members

Set $A \cap C$ and set $B \cap C$ have no members in common.

(c) List the 4 members of set C

(Total for Question 4 is 5 marks)

Q12**PAST PAPER**

Edexcel 4MA1 • May 2024 • Paper 1H • Q7

CH 1 • TOPIC 2 5 marks

7 $\mathcal{E} = \{23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34\}$

$A = \{\text{even numbers}\}$

$B = \{23, 29, 31\}$

$C = \{\text{multiples of 3}\}$

(a) List the members of the set

(i) $B \cup C$

.....

(ii) $A' \cap C$

.....

(b) Is it true that $B \cap C = \emptyset$?

Tick (✓) one of the boxes below.

Yes

No

☐
☐

Give a reason for your answer.

.....

The set D has 4 members and is such that $D \cap (A \cup C) = \emptyset$

(c) List the members of set D

.....

(Total for Question 7 is 5 marks)

Q13**PAST PAPER**

Edexcel 4MA1 • June 2025 • Paper 1HR • Q5

CH 1 • TOPIC 2 4 marks

5 $\mathcal{E} = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12\}$

$$A = \{2, 4, 6, 8, 10, 12\}$$

$$B = \{3, 6, 9, 12\}$$

$$C = \{1, 3, 5, 7, 9, 11\}$$

(a) List the members of the set

(i) $A \cup B$

.....

(ii) B'

.....

$$\mathcal{E} = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12\}$$

$$A = \{2, 4, 6, 8, 10, 12\}$$

$$B = \{3, 6, 9, 12\}$$

$$C = \{1, 3, 5, 7, 9, 11\}$$

| $\mathcal{E}$ $\cap$ $\cup$ $\emptyset$ $\in$ $\notin$ |

(b) Write a symbol from the box on each dotted line to make each of the following a true statement.

(i) $A \cap C = \dots\dots\dots$

(ii) $13 \dots\dots\dots \mathcal{E}$

(Total for Question 5 is 4 marks)

02 Read a Venn Diagram

Numbers in a region can mean two things: the **members** themselves, or **how many** members (n). Read the question: “list” or “ $n(\dots)$ ”.

WHEN YOU SEE → YOU DO

“list the members of A ”

→ every number inside circle A — including the overlap

“ $n(A \cup B)$ ”

→ add every region inside A or B

“ $n(A' \cap B')$ ”

→ outside both circles

“represent the numbers of elements”

→ add the numbers — do not list them

WORKED MODEL • MAY 2021 P1H

Regions: A only 12, $A \cap B$ only 5, B only 9, all three 6, $A \cap C$ only 10, $B \cap C$ only 3, C only 4, outside 8. Find $n(A \cup B)$ and $n(A' \cap B')$.

1 $12 + 5 + 9 + 6 + 10 + 3 = 45$

every region in A or B

2 $4 + 8 = 12$

outside A and outside B

ANSWER $n(A \cup B) = 45$, $n(A' \cap B') = 12$

IDEA 02

Practise: Read a Venn Diagram

My questions first, then selected practice, then real past papers.

Q14 DR ESLAM Count, do not list

CH 1 • TOPIC 2 2 marks

A Venn diagram has A only 7, $A \cap B$ 4, B only 9, outside 5. Find $n(A \cup B)$ and $n(B')$.

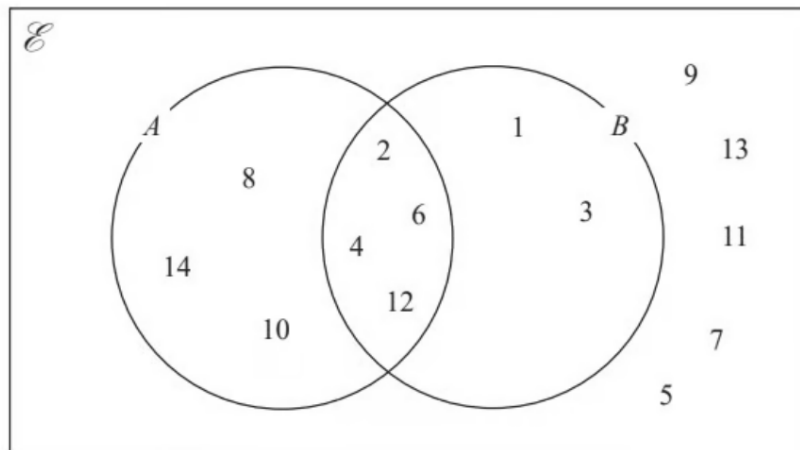
SELECTED PRACTICE

Q15**PRACTICE**

Practice • Easy

CH 1 • TOPIC 2 2 marks

10 The numbers from 1 to 14 are shown in the Venn diagram.



i) List the members of the set $A \cap B$

[1]

ii) List the members of the set B'

[1]

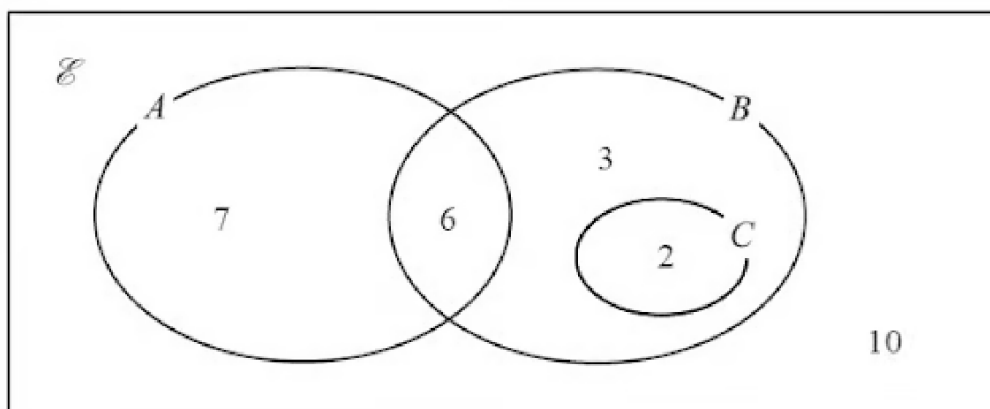
(2 marks)

Q16**PRACTICE**

Practice • Medium

CH 1 • TOPIC 2 4 marks

6 (a) The Venn diagram shows a universal set ξ and three sets A , B and C .



7, 6, 3, 2 and 10 represent the **number** of elements in each set.

Find

$$n(A \cup B)$$

(1 mark)

(b) $n(A')$

(1 mark)

(c) $n(B \cap C)$

(1 mark)

(d) $n(A' \cup B')$

(1 mark)

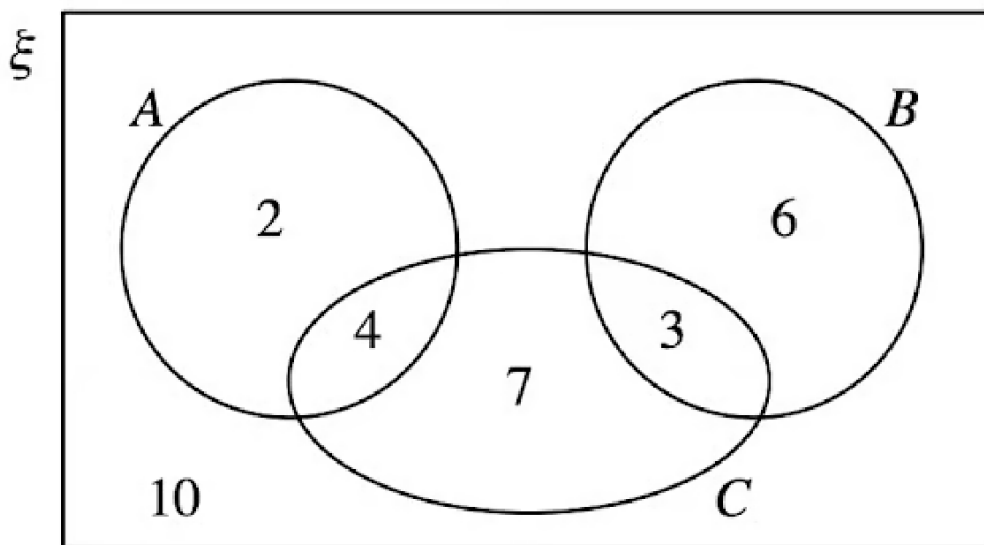
Q17**PRACTICE**

Practice • Hard

CH 1 • TOPIC 2

4 marks

5 (a) The Venn diagram shows a universal set ξ and three sets A , B and C .



2, 4, 7, 3, 6 and 10 represent the **number** of elements in each set.

Find

$$n(A \cap B)$$

(1 mark)

(b) $n(B')$

(1 mark)

(c) $n(A \cap C)$

(1 mark)

(d) $n(B' \cap C)$

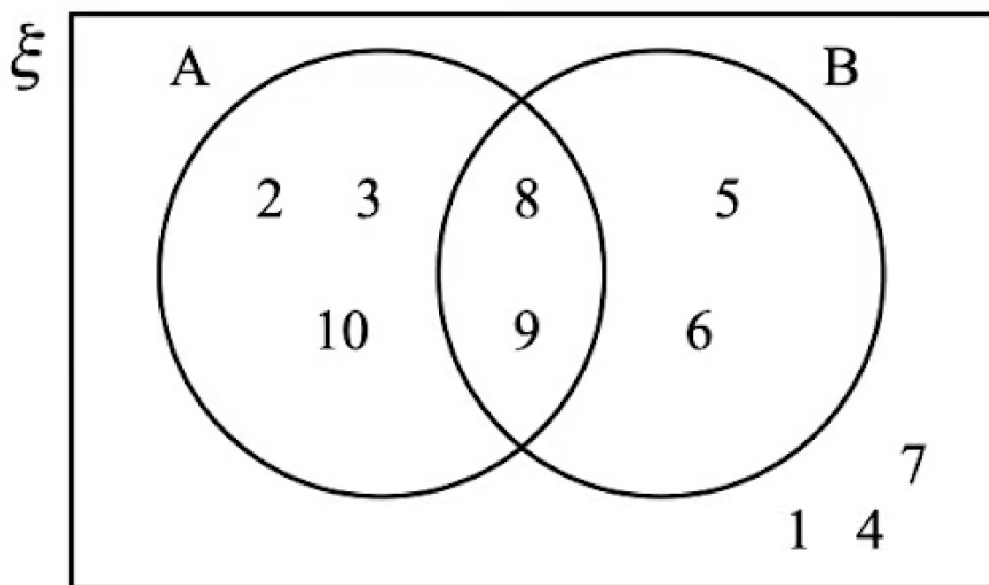
(1 mark)

Q18**PRACTICE**

Practice • Hard

CH 1 • TOPIC 2 4 marks

6 (a) The Venn diagram shows all of the elements in sets A , B and ξ .



Write down the elements in A' ,

(1 mark)

(b) Find $n(A \cap B)'$,

(1 mark)

(c) Find the elements in $(A \cap B) \cup (A \cup B)'$,

(1 mark)

(d) $A \cap C = \emptyset$

$$B \cup C = \{5, 6, 7, 8, 9\}$$

$$n(C) = 3$$

Write down the elements in C .

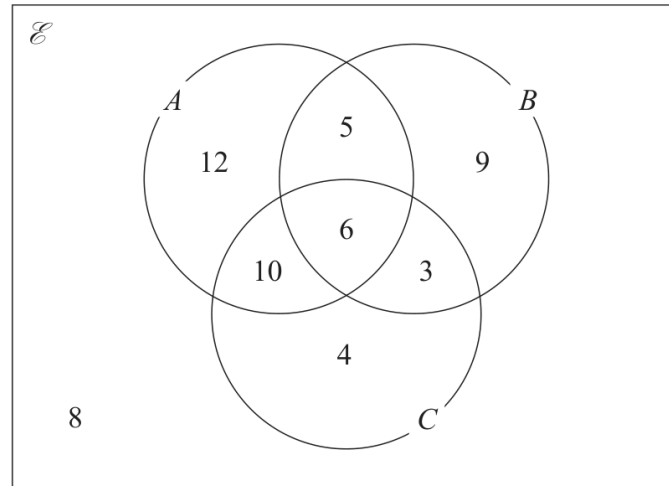
(1 mark)

PAST PAPERS

Q19 PAST PAPER Edexcel 4MA1 • May 2021 • Paper 1H • Q19

CH 1 • TOPIC 2 3 marks

19 The Venn diagram shows a universal set, $\mathcal{E}$ and sets A , B and C .



12, 5, 9, 10, 6, 3, 4 and 8 represent the **numbers** of elements.

Find

(i) $n(A \cup B)$

.....

(ii) $n(A' \cap B')$

.....

(iii) $n([A \cap B] \cup C)$

.....

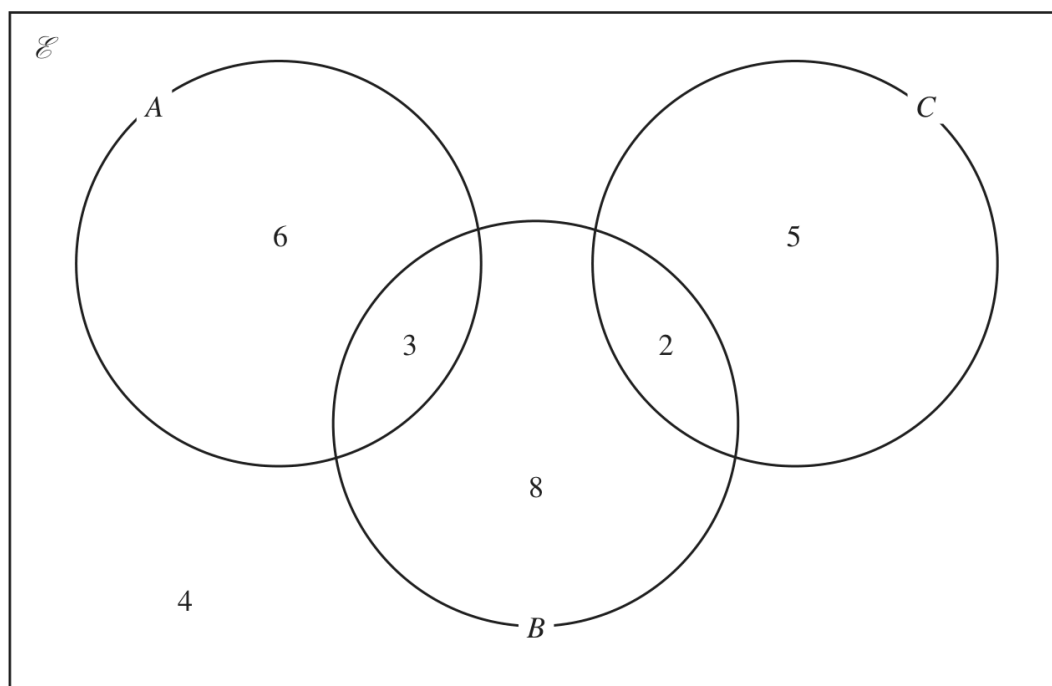
(Total for Question 19 is 3 marks)

Q20**PAST PAPER**

Edexcel 4MA1 • January 2022 • Paper 1HR • Q17

CH 1 • TOPIC 2 4 marks

17 The Venn diagram shows a universal set $\mathcal{E}$ and three sets A , B and C .



6, 3, 8, 2, 5 and 4 represent the **numbers** of elements.

Find

(i) $n(A \cup B)$

.....

(ii) $n(A \cap C)$

.....

(iii) $n(B \cap C')$

.....

(iv) $n(A' \cup B' \cup C')$

.....

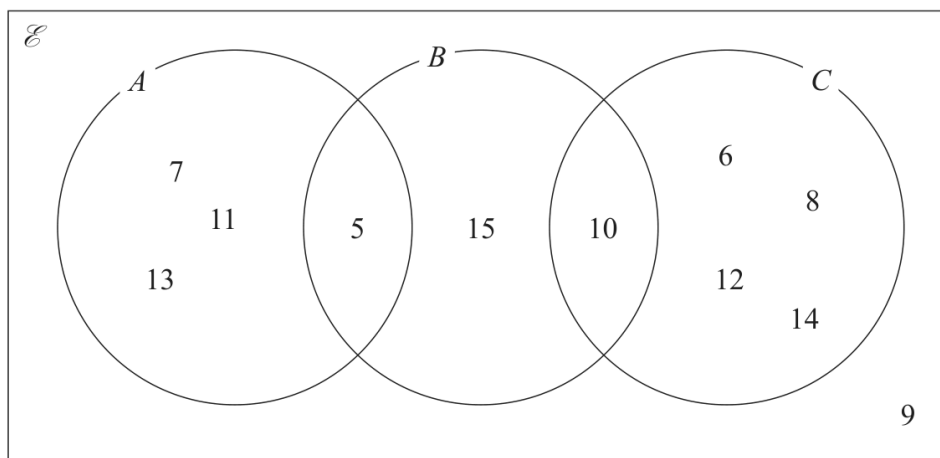
(Total for Question 17 is 4 marks)

Q21**PAST PAPER**

Edexcel 4MA1 • May 2023 • Paper 1HR • Q2

CH 1 • TOPIC 2 3 marks

2 Here is a Venn diagram.



(a) Write down the numbers that are in the set

(i) A

.....

(ii) $B \cup C$

.....

Dominic writes down $9 \notin C$

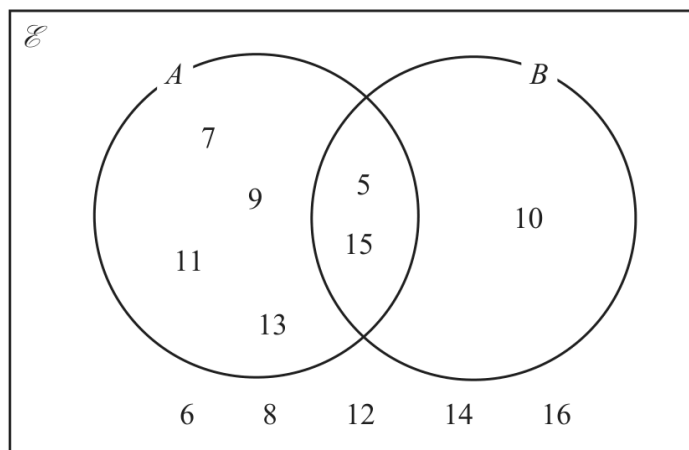
(b) Explain why Dominic is correct.

.....

(Total for Question 2 is 3 marks)

Q22**PAST PAPER**

Edexcel 4MA1 • November 2023 • Paper 1H • Q1

CH 1 • TOPIC 2 3 marks**You must write down all the stages in your working.****1** Here is a Venn diagram.

List the members of the set

(a) A

.....

(b) $A \cap B$

.....

(c) $(A \cup B)'$

.....

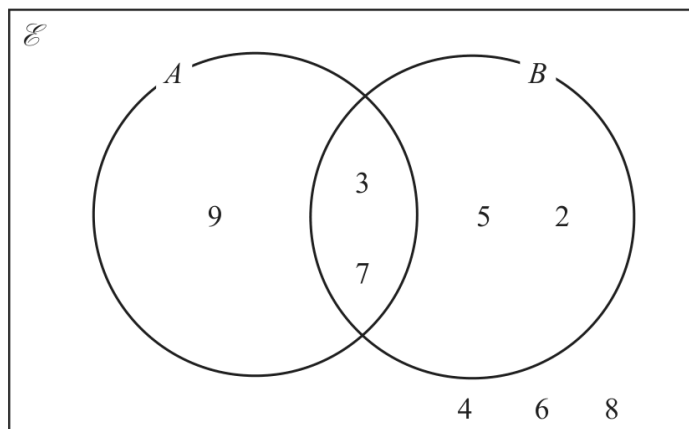
(Total for Question 1 is 3 marks)

Q23**PAST PAPER**

Edexcel 4MA1 • November 2024 • Paper 1H • Q4

CH 1 • TOPIC 2 3 marks

4 Here is a Venn diagram.



(a) List the members of the set B

.....

(b) List the members of the set $A \cap B$

.....

(c) List the members of the set A'

.....

(Total for Question 4 is 3 marks)

03 Shade a Region

Shade in stages: first the bracket, then the symbol outside it. Describe a shaded region by naming the circles it is **in** and **not in**.

WHEN YOU SEE → YOU DO

“shade $(A \cup B) \cap C$ ”

→ shade $A \cup B$, then remove C

“ $C \subset A$ ”

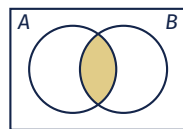
→ circle C drawn completely inside A

“ $A \cap B = \emptyset$ ”

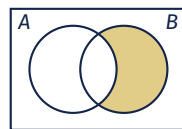
→ circles do not overlap

“use set notation to describe the shaded region”

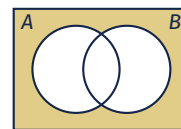
→ list what it is in, what it is not in



$A \cap B$



$A' \cap B$ (B only)



$A' \cap B' = (A \cup B)'$

TWO SYMBOLS $\subset$ = “is inside” (a subset). $\in$ = “is a member of”, $\notin$ = “is not a member of”.

IDEA 03

Practise: Shade a Region

My questions first, then selected practice.

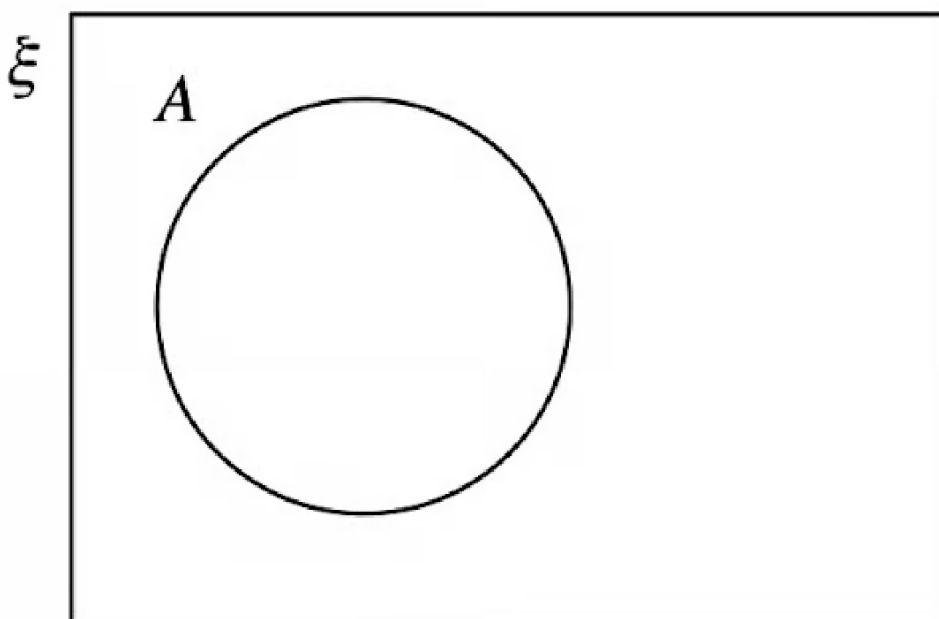
SELECTED PRACTICE**Q24****PRACTICE**

Practice • Medium

CH 1 • TOPIC 2 3 marks

5 (a) A , B and C are three sets.

$$A \cap B = \emptyset \text{ and } C \subset A$$

Complete the Venn diagram to show the sets B and C .**(2 marks)****(b)** On the Venn diagram, shade the region that represents $A \cap C$.**(1 mark)**

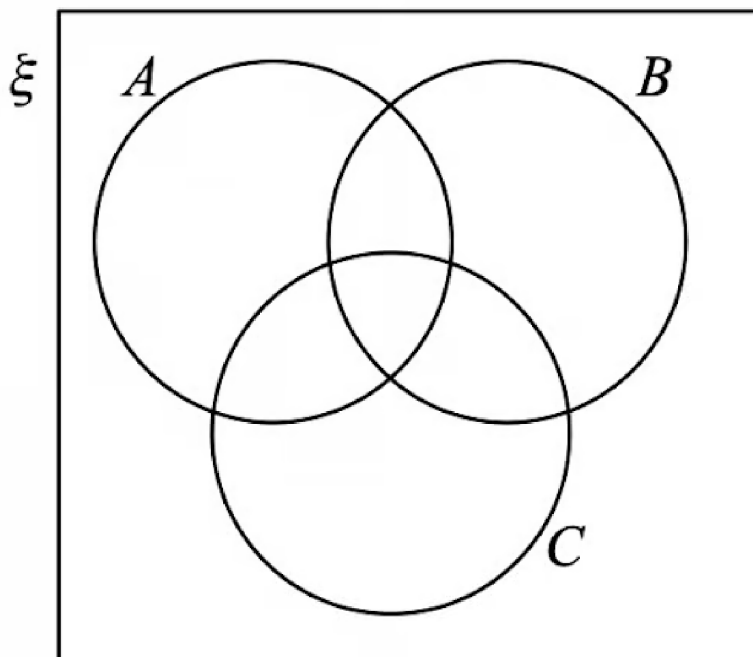
Q25

PRACTICE

Practice • Hard

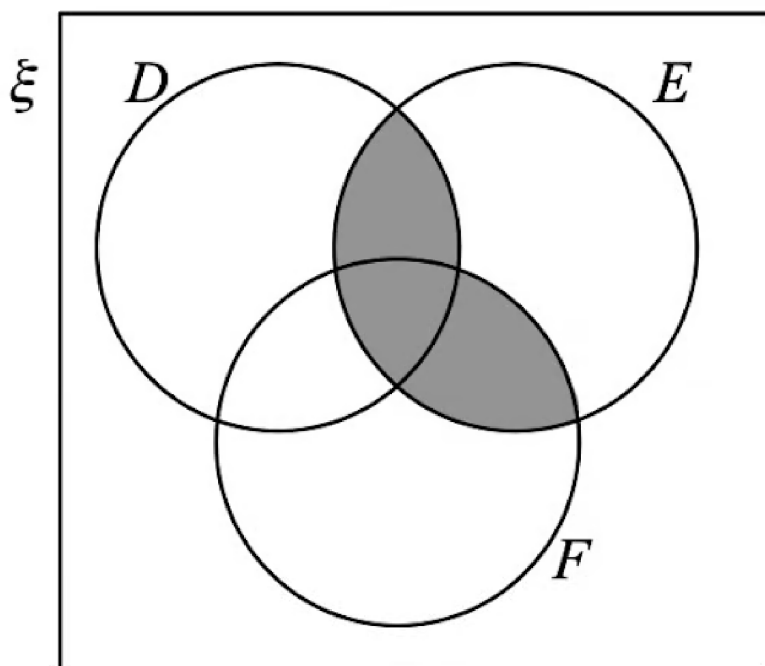
CH 1 • TOPIC 2 4 marks

7 (a) On the Venn diagram, shade the set $(A \cup B) \cap C'$.



(2 marks)

(b) Use set notation to describe the shaded region in the Venn diagram below.



(2 marks)

04 Fill It In: Work from the Middle

Always fill the **overlap first**, then the “only” parts, then the outside. Check that everything adds to the total.

WHEN YOU SEE → YOU DO

“ $n(A \cap B) = 12$ ”

→ write 12 in the overlap first

“ $n(A) = 22$ ”

→ $A \text{ only} = 22 - 12$

“6 have neither”

→ goes outside the circles, inside the box

“how many study only one language?”

→ add the two “only” regions

given $A \cap B, A', B'$ as lists

→ place each member in its one region

Overlap

Only A and only B

Outside

Check the total

WORKED MODEL • TWO SETS

50 people: 30 own a cat, 25 own a dog, 6 own neither. Complete the Venn diagram.

1 $30 + 25 - x + 6 = 50 \Rightarrow x = 11$

overlap x

2 cat only $30 - 11 = 19$, dog only $25 - 11 = 14$

“only” parts

ANSWER 19, 11, 14 and 6 outside

× **TRAP** Writing 30 in the “cat only” region. 30 is the whole circle — the overlap is part of it.

IDEA 04

Practise: Fill It In: Work from the Middle

My questions first, then selected practice, then real past papers.

Q26 DR ESLAM Middle first

CH 1 • TOPIC 2 3 marks

40 students: 22 play football, 18 play tennis, 7 play both. How many play neither?

SELECTED PRACTICE

Q27**PRACTICE**

Practice • Easy

CH 1 • TOPIC 2 **3 marks**

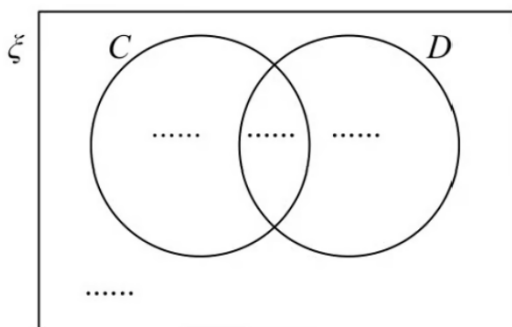
5 In a survey of 50 people,

30 owned a cat

25 owned a dog

6 owned neither a cat or a dog

Complete the Venn diagram to show this information.

**(3 marks)**

Q28**PRACTICE**

Practice • Medium

CH 1 • TOPIC 2 5 marks

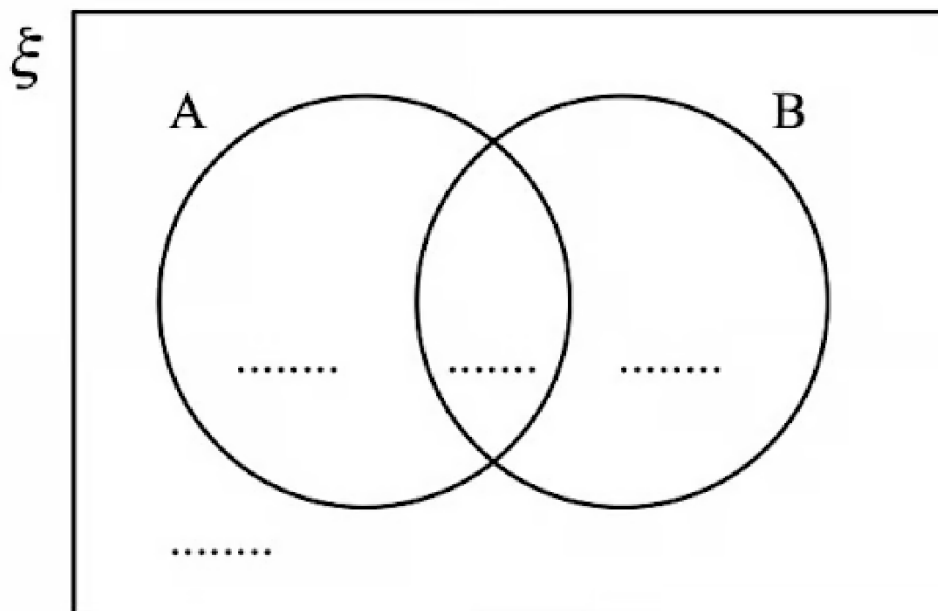
11 (a) A and B are two sets.

$$n(\xi) = 36$$

$$n(B) = 21$$

$$n(A \cap B) = 8$$

$$n(A') = 18$$

Complete the Venn diagram to show the **number of elements** in each region.**(3 marks)****(b)** Find

i) $n(A \cup B)$

ii) $n(A \cap B')$

(2 marks)

Q29**PRACTICE**

Practice • Hard

CH 1 • TOPIC 2 4 marks

2 There are 31 students in a class.

The only languages available for the class to study are French and Spanish.

17 students study French.

15 students study Spanish.

6 students study neither French nor Spanish.

Using a Venn diagram, or otherwise, work out how many students study only one language.

(4 marks)

Q30**PRACTICE**

Practice • Hard

CH 1 • TOPIC 2 5 marks

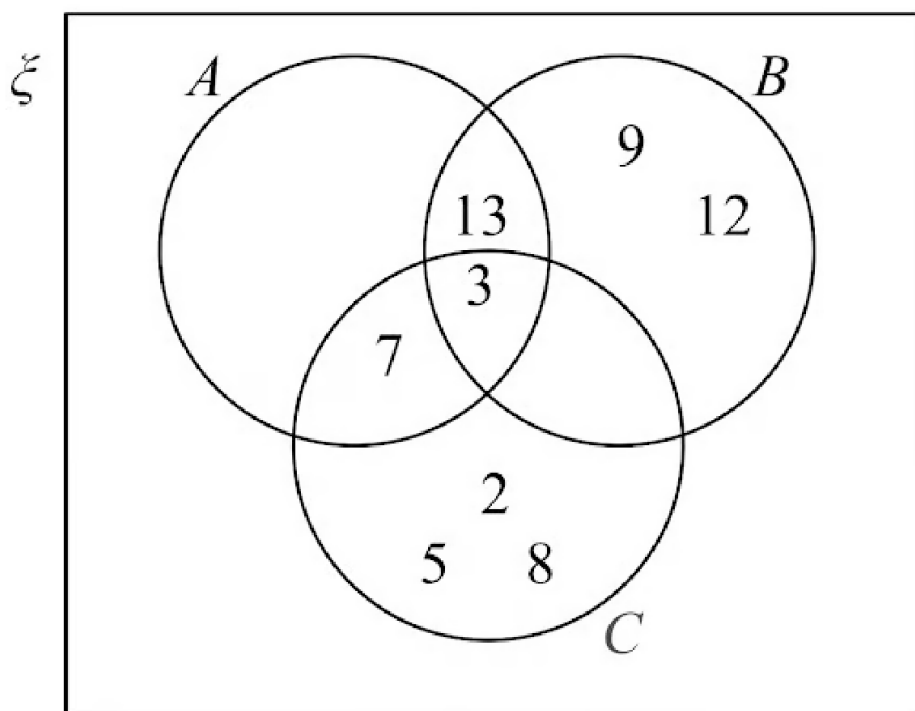
8 (a) $\xi = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13\}$

$$A = \{3, 7, 11, 13\}$$

$$B = \{3, 6, 9, 12, 13\}$$

$$C = \{2, 3, 5, 6, 7, 8\}$$

Complete the Venn diagram



(2 marks)

(b) List the members of the set $B' \cap C$.

(1 mark)

(c) List the members of the set $(A \cup C)' \cap B$.

(1 mark)

(d) Find $n(A' \cap B')$

(1 mark)

PAST PAPERS

Q31 PAST PAPER Edexcel 4MA1 • January 2022 • Paper 1H • Q7

CH 1 • TOPIC 2 3 marks

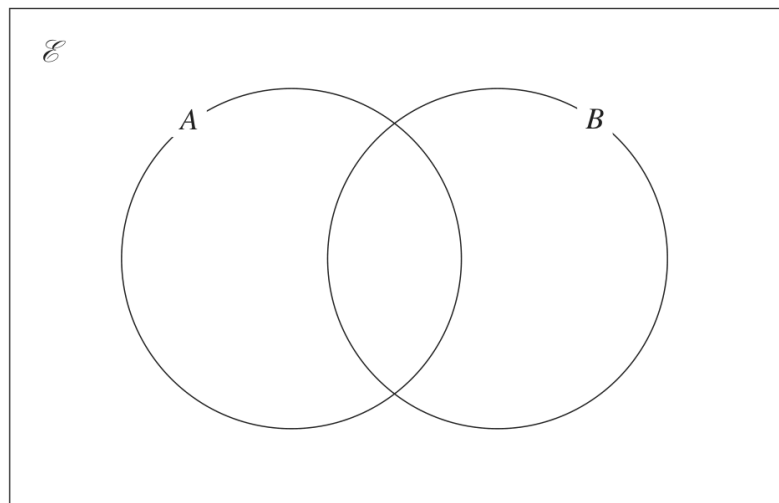
7 $\mathcal{E} = \{4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15\}$

$$A \cap B = \{5, 10, 15\}$$

$$B' = \{7, 8, 9, 11, 12, 13, 14\}$$

$$A' = \{4, 6, 7, 8, 14\}$$

Complete the Venn diagram for this information.



(Total for Question 7 is 3 marks)

05 Three Sets with x

Same rule with three circles: the centre first, then the three pair-regions, then the “only” regions — in terms of x if needed. Then use the total.

WHEN YOU SEE → YOU DO

“6 have a dog, a cat and a rabbit”

→ centre

“9 have a dog and a rabbit”

→ pair region = 9 – centre

“Let x be the number who ...”

→ write every other region in terms of x

“given the total is 50, find x ”

→ add every region = 50, solve

Centre

Pairs minus centre

Only regions

Outside

Total = equation

WORKED MODEL • JANUARY 2020 P2HR

Dog 28, cat 18, rabbit 20; cat & rabbit 8, dog & rabbit 9, dog & cat x ; all three 6; none 5; total 50.

1 centre 6; C&R only 2, D&R only 3, D&C only $x - 6$

pairs

2 R only 9, D only $25 - x$, C only $16 - x$

only regions

3 $60 - x = 50$

total

ANSWER $x = 10$

WHERE YOU WILL USE IT Probability from Venn diagrams (Chapter 6): the same regions become probabilities.

IDEA 05

Practise: Three Sets with x

My questions first, then selected practice, then real past papers.

Q32 DR ESLAM Three sets

CH 1 • TOPIC 2 4 marks

60 people. All three sports: 4. $A \cap B$ only: 6, $B \cap C$ only: 5, $A \cap C$ only: 3. A only: x , B only: $2x$, C only: 10. None: 8. Find x .

$x =$ _____

SELECTED PRACTICE

Q33**PRACTICE**

Practice • Hard

CH 1 • TOPIC 2

4 marks

3 A garage tests cars for faults.

There are three types of fault – braking, steering and lighting.

A car fails the test if it has one or more of these three types of fault.

Last week, 11 cars had breaking faults,

9 cars had steering faults,

7 cars had lighting faults,

no car had both steering faults and lighting faults,

2 cars had both braking faults and steering faults,

3 cars had both braking faults and lighting faults.

By drawing a Venn diagram, or otherwise, find the number of cars which failed the test last week.

(4 marks)

Q34**PRACTICE**

Practice • Hard

CH 1 • TOPIC 2 **3 marks**

- 9 Each student in a group of 32 students was asked the following question.

“Do you have a desktop computer (D), a laptop (L) or a tablet (T)?”

Their answers showed that,

19 students have a desktop computer,

17 students have a laptop,

16 students have a tablet,

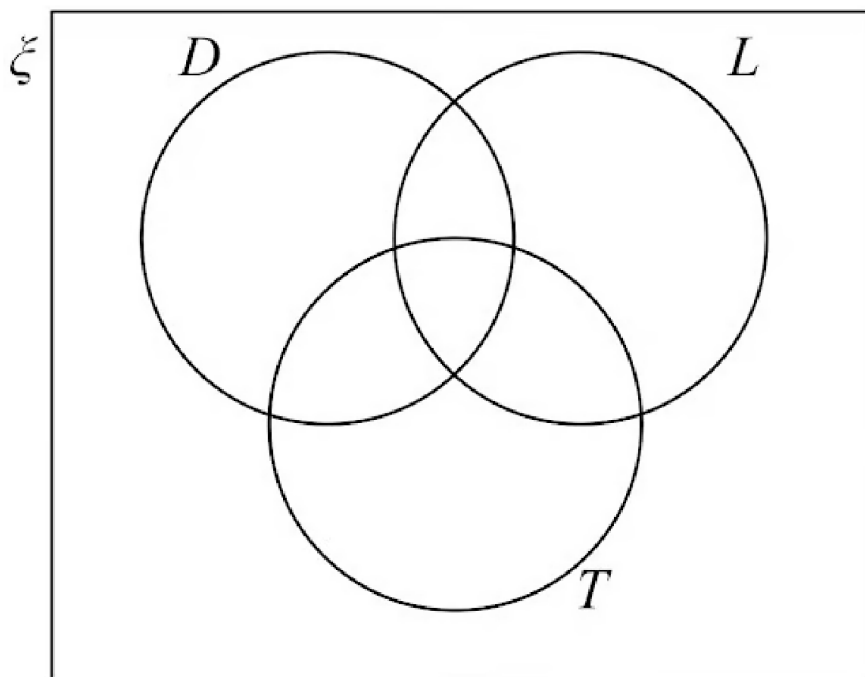
9 students have both a desktop computer and a laptop,

11 students have both a desktop computer and a tablet,

7 students have both a laptop and a tablet,

5 students have all three.

Using this information, complete the Venn diagram to show the number of students in each appropriate subset.

**(3 marks)**

Q35**PRACTICE**

Practice • Hard

CH 1 • TOPIC 2 **3 marks**

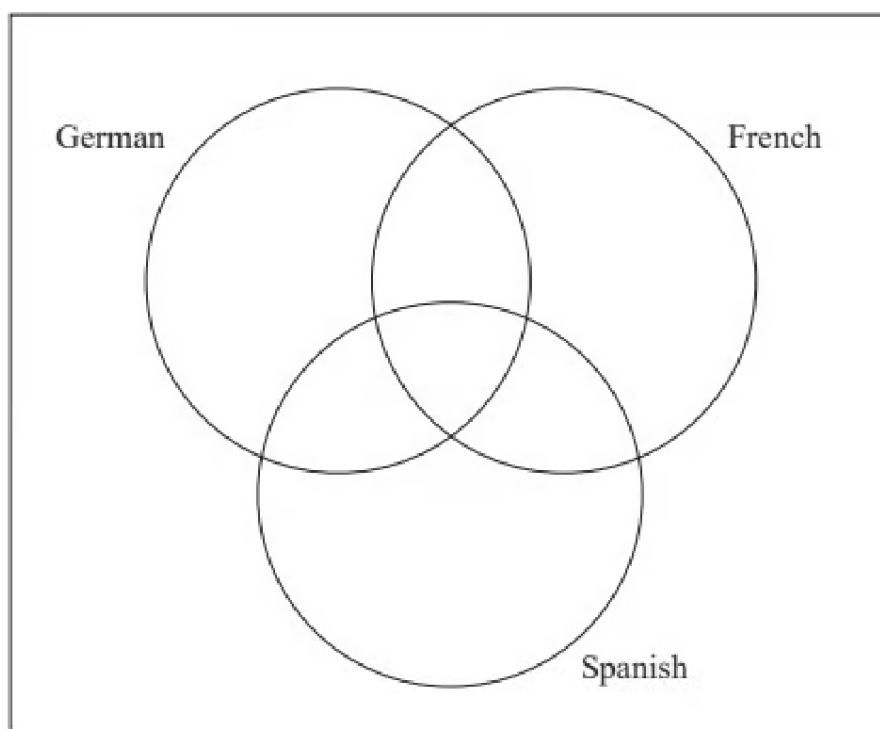
- 13** In a school, students must study at least one language from German, French and Spanish.

There are 30 students in a class of this school.

Of these students

- 7 study all three of the languages
- 10 study both German and French
- 12 study both Spanish and German
- 9 study both French and Spanish
- 16 study Spanish
- 18 study German

Work out the total number of the students in the class who study French.
You may use the Venn diagram to help with your calculations.

**(3 marks)**

PAST PAPERS

Q36 PAST PAPER Edexcel 4MA1 • January 2020 • Paper 2HR • Q17

CH 1 • TOPIC 2 6 marks

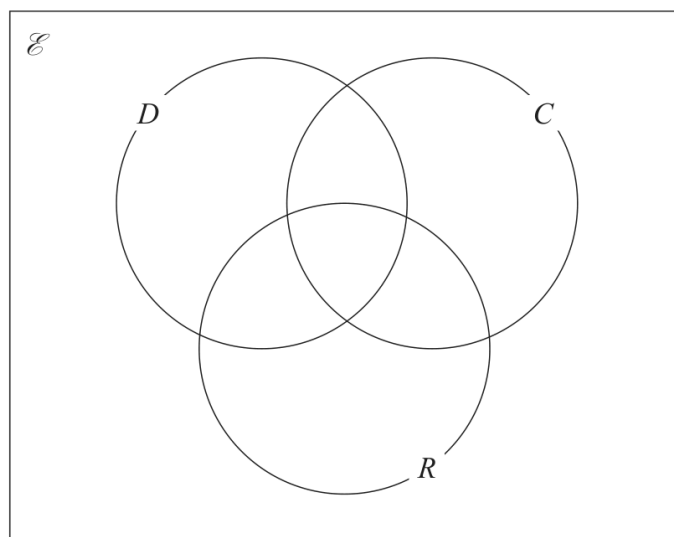
17 Some students in a school were asked the following question.

“Do you have a dog (D), a cat (C) or a rabbit (R)?”

Of these students

- 28 have a dog
- 18 have a cat
- 20 have a rabbit
- 8 have both a cat and a rabbit
- 9 have both a dog and a rabbit
- x have both a dog and a cat
- 6 have a dog, a cat and a rabbit
- 5 have not got a dog or a cat or a rabbit

- (a) Using this information, complete the Venn diagram to show the number of students in each appropriate subset.
Give the numbers in terms of x where necessary.



Given that a total of 50 students answered the question,

- (b) work out the value of x .

$x =$
(2)

- (c) Find $n(C' \cap D')$

.....

(Total for Question 17 is 6 marks)

Q37**PAST PAPER**

Edexcel 4MA1 • January 2021 • Paper 2H • Q16

CH 1 • TOPIC 2 6 marks

16 Some students were asked the following question.

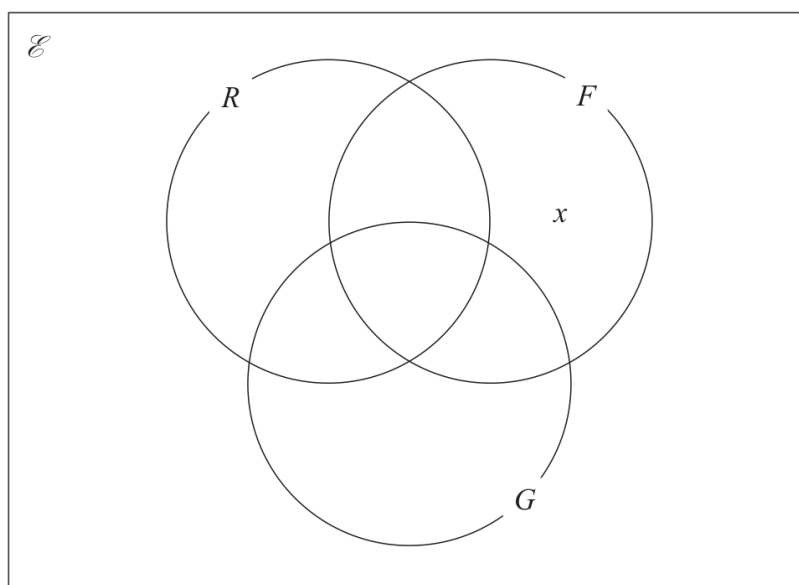
“Which of the subjects Russian (R), French (F) and German (G) do you study?”

Of these students

- 4 study all three of Russian, French and German
- 10 study Russian and French
- 13 study French and German
- 6 study Russian and German
- 24 study German
- 11 study none of the three subjects
- the number who study Russian only is twice the number who study French only.

Let x be the number of students who study French only.

- (a) Show all this information on the Venn diagram, giving the number of students in each appropriate subset, in terms of x where necessary.



Given that the number of students who were asked the question was 80

- (b) work out the number of these students that study Russian.

(Total for Question 16 is 6 marks)

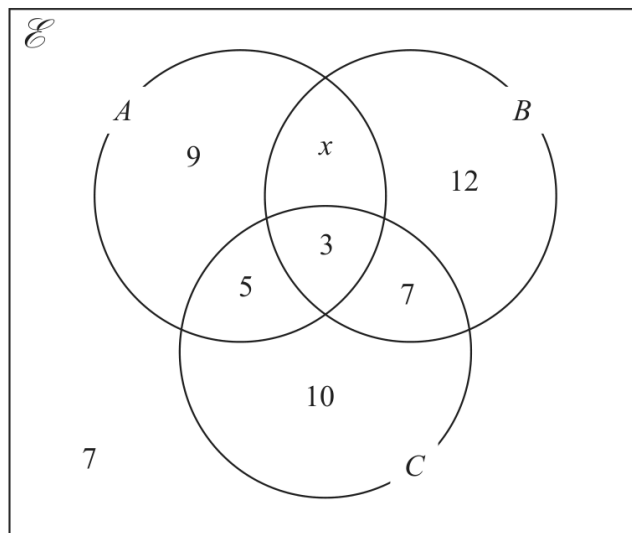
Q38**PAST PAPER**

Edexcel 4MA1 • May 2023 • Paper 1H • Q16

CH 1 • TOPIC 2 3 marks

16 The Venn diagram shows a universal set $\mathcal{E}$ and sets A , B and C

The numbers and the letter x represent **numbers** of elements.



Given that $n(A \cup B) = 42$

(a) find the value of x

$x = \dots\dots\dots$

(b) Find $n(A')$

$\dots\dots\dots$

(c) Find $n(B' \cap C)$

$\dots\dots\dots$

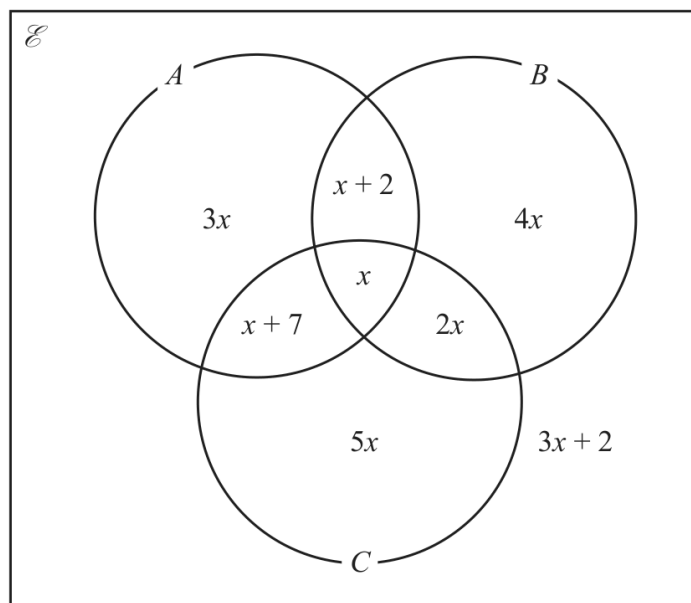
(Total for Question 16 is 3 marks)

Q39**PAST PAPER**

Edexcel 4MA1 • November 2024 • Paper 2H • Q21

CH 1 • TOPIC 2 4 marks

- 21 The Venn diagram shows information about the numbers of items in set A , set B and set C , where x is an integer.



Given that $n(A \cup B)' = 26$

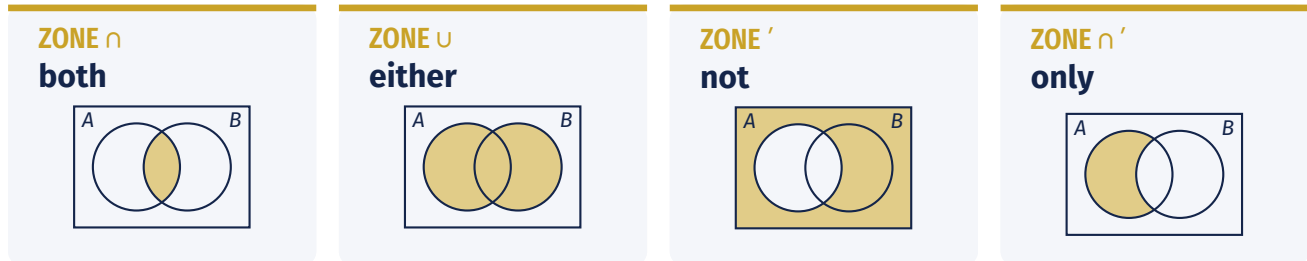
find $n(A' \cap C)$

$n(A' \cap C) = \dots\dots\dots$

(Total for Question 21 is 4 marks)

ALL The Whole Topic on One Page

Tear this page out before the exam.



- | | | |
|-----------|----------------------|---|
| 01 | Notation | Write E, A, B out; apply the symbol; each member once. |
| 02 | Read | “List” = the members; “ $n()$ ” = add the numbers. |
| 03 | Shade | Bracket first. Describe: in what, not in what. |
| 04 | Fill (2 sets) | Overlap $\rightarrow$ only $\rightarrow$ outside $\rightarrow$ check the total. |
| 05 | Fill (3 sets) | Centre $\rightarrow$ pairs $\rightarrow$ only $\rightarrow$ outside $\rightarrow$ total = equation in x . |

BOTTOM LINE Always start in the middle of the Venn diagram $\rightarrow$ work outwards $\rightarrow$ check the total.

Answers

Set Notation and Venn Diagrams · for the teacher. Every answer checked with a computer algebra system.

IDEA 01 · SET NOTATION: LIST THE MEMBERS

- Q1** Dr Eslam • Both and either (a) $\{2, 4, 6, 12\}$ (b) $\{5, 7, 8, 9, 10, 11\}$
- Q2** Practice • Easy (a)(i) $\{p, r, a\}$ (ii) $\{p, a, r, i, s, b, u, d, e, t\}$ (b) E : no letter of A is in E
- Q3** Practice • Easy (a) no member of A is in B (b) 1, 9 (c) $\{1, 2, 8, 9\}$
- Q4** Practice • Medium $\{5, 10, 20, 25, 50, 100\}$
- Q5** Practice • Hard (a)(i) $\{1, 5, 7, 11\}$ (ii) $\{2, 3, 5, 6, 7, 9, 11, 12\}$ (b) No: 14 is not in E
- Q6** Edexcel 4MA1 • January 2020 • Paper 1HR • Q4 (a)(i) $\{b, l, u, e, g, r, y\}$ (ii) $\{w, h, i, t\}$ (b) No: e is in all three
- Q7** Edexcel 4MA1 • January 2021 • Paper 1H • Q4 (i) $\{21, 27\}$ (ii) $\{21, 23, 24, 25, 27, 29\}$
- Q8** Edexcel 4MA1 • January 2021 • Paper 2HR • Q3 (a)(i) $\{24, 30\}$ (ii) $\{21, 23, 25, 27, 29\}$ (b) $C = (A \cup B)'$
- Q9** Edexcel 4MA1 • May 2021 • Paper 2H • Q4 (a)(i) $\{b, r, a, z, i, l, e, n, d\}$ (ii) $\{b, z\}$ (b) a is in both B and K
- Q10** Edexcel 4MA1 • June 2022 • Paper 2HR • Q4 (a)(i) $\{9, 15\}$ (ii) $\{9, 11, 12, 13, 15, 17, 18, 19\}$ (b) No: 24 is not in E (c) e.g. $\{9, 10, 11, 18\}$
- Q11** Edexcel 4MA1 • June 2024 • Paper 2HR • Q4 (a)(i) $\{1, 2, 3, 5, 6, 7\}$ (ii) $\{4, 5, 7, 8, 9, 10\}$ (b) 2 and 3 are in both (c) $\{1, 5, 6, 7\}$
- Q12** Edexcel 4MA1 • May 2024 • Paper 1H • Q7 (a)(i) $\{23, 24, 27, 29, 30, 31, 33\}$ (ii) $\{27, 33\}$ (b) Yes: none of 23, 29, 31 is a multiple of 3 (c) $\{23, 25, 29, 31\}$
- Q13** Edexcel 4MA1 • June 2025 • Paper 1HR • Q5 (a)(i) $\{2, 3, 4, 6, 8, 9, 10, 12\}$ (ii) $\{1, 2, 4, 5, 7, 8, 10, 11\}$ (b)(i) $\emptyset$ (ii) $\notin$

IDEA 02 • READ A VENN DIAGRAM

- Q14** Dr Eslam • Count, do not list $n(A \cup B) = 20, n(B') = 12$
- Q15** Practice • Easy (i) $\{2, 4, 6, 12\}$ (ii) $\{5, 7, 8, 9, 10, 11, 13, 14\}$
- Q16** Practice • Medium (a) 18 (b) 15 (c) 9 (d) 22
- Q17** Practice • Hard (a) 0 (b) 23 (c) 2 (d) 12
- Q18** Practice • Hard (a) $\{1, 4, 5, 6, 7\}$ (b) 8 (c) $\{1, 4, 7, 8, 9\}$ (d) $\{5, 6, 7\}$
- Q19** Edexcel 4MA1 • May 2021 • Paper 1H • Q19 (i) 45 (ii) 12 (iii) 28
- Q20** Edexcel 4MA1 • January 2022 • Paper 1HR • Q17 (i) 19 (ii) 0 (iii) 11 (iv) 28
- Q21** Edexcel 4MA1 • May 2023 • Paper 1HR • Q2 (a)(i) $\{5, 7, 11, 13\}$ (ii) $\{5, 6, 8, 10, 12, 14, 15\}$ (b) 9 is outside circle C
- Q22** Edexcel 4MA1 • November 2023 • Paper 1H • Q1 (a) $\{5, 7, 9, 11, 13, 15\}$ (b) $\{5, 15\}$ (c) $\{6, 8, 12, 14, 16\}$
- Q23** Edexcel 4MA1 • November 2024 • Paper 1H • Q4 (a) $\{2, 3, 5, 7\}$ (b) $\{3, 7\}$ (c) $\{2, 4, 5, 6, 8\}$

IDEA 03 • SHADE A REGION

- Q24** Practice • Medium (a) B separate from A; C inside A (b) A outside C shaded
- Q25** Practice • Hard (a) $A \cup B$ outside C shaded (b) $E \cap (D \cup F)$

IDEA 04 • FILL IT IN: WORK FROM THE MIDDLE

- Q26** Dr Eslam • Middle first $15 + 7 + 11 = 33$; neither = 7
- Q27** Practice • Easy cat only 19, both 11, dog only 14, outside 6
- Q28** Practice • Medium (a) 10, 8, 13, outside 5 (b)(i) 31 (ii) 10
- Q29** Practice • Hard both = $17 + 15 - 25 = 7$; only one = 18
- Q30** Practice • Hard (a) 11 in A only, 6 in $B \cap C$ only, 1, 4, 10 outside
(b) {2, 5, 7, 8} (c) {9, 12} (d) 6
- Q31** Edexcel 4MA1 • January 2022 • Paper 1H • Q7
A only 9, 11, 12, 13; $A \cap B$ 5, 10, 15; B only 4, 6;
outside 7, 8, 14

IDEA 05 • THREE SETS WITH x

- Q32** Dr Eslam • Three sets $3x + 36 = 60$; $x = 8$
- Q33** Practice • Hard braking only 6, B&S 2, B&L 3, steering only 7,
lighting only 4: 22 cars
- Q34** Practice • Hard centre 5; D&L 4, D&T 6, L&T 2; D only 4, L only
6, T only 3; outside 2
- Q35** Practice • Hard French only 8: French total 20
- Q36** Edexcel 4MA1 • January 2020 • Paper 2HR • Q17
(b) $x = 10$ (c) 14
- Q37** Edexcel 4MA1 • January 2021 • Paper 2H • Q16
(a) R only $2x$, F only x , R&F only 6, all 4, R&G
only 2, F&G only 9, G only 9, none 11 (b) $x =$
13: 38 study Russian
- Q38** Edexcel 4MA1 • May 2023 • Paper 1H • Q16
(a) $x = 6$ (b) 36 (c) 15
- Q39** Edexcel 4MA1 • November 2024 • Paper 2H • Q21
 $8x + 2 = 26$, $x = 3$: $n(A' \cap C) = 7x = 21$